

Project Design Phase-II Technology Stack (Architecture & Stack)

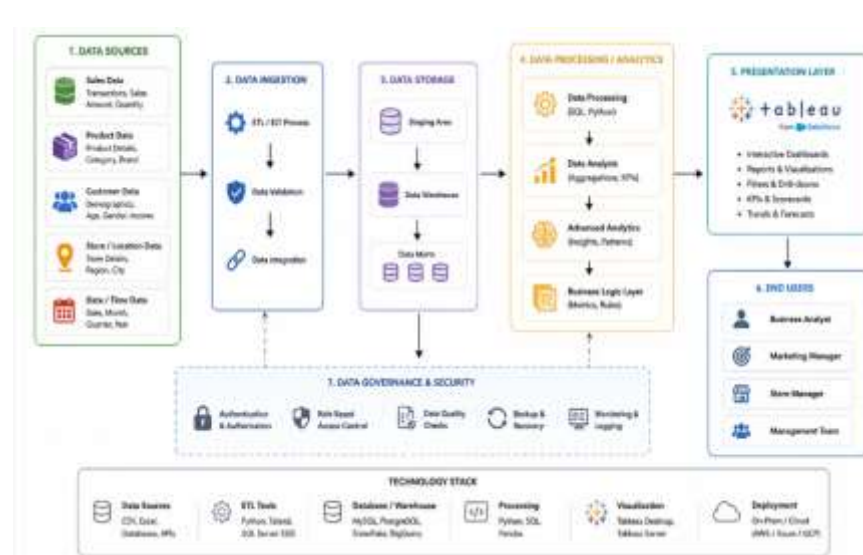
Date	4 July 2026
Team ID	
Project Name	Strategic product placement analysis
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive dashboards for viewing sales and placement insights	Tableau public, Tableau Desktop
2.	Application Logic-1	Data Cleaning and preprocessing	Tableau public
3.	Application Logic-2	Sales and consumer behaviour analysis	Tableau public
4.	Application Logic-3	Product placement analysis and KPI calculations	Tableau public
5.	Database	Data Type, Configurations etc.	None
6.	Cloud Database	Database Service on Cloud	Tableau
7.	File Storage	File storage requirements	CSV file/Excel file
8.	External API-1	Purpose of External API used in the application	None
9.	External API-2	Purpose of External API used in the application	None
10.	Machine Learning Model	Purpose of Machine Learning Model	None
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	None

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	None
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	None

S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	None
4.	Availability	Keep dashboards accessible and backed up	Tableau Public
5.	Performance	Fast data retrieval	Tableau

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>